🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.NOS.4

Least Common Multiple

Lesson 6-12 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the least common multiple (LCM) of two numbers by listing or comparing their multiples.
Language Objective: I can explain how I found the LCM using the words multiple, common multiple, skip counting, and least common multiple.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Least Common Multiple | Multiple | Common multiple | Skip counting | Prime factorization

👆 *Tap any word to see its meaning and a picture.*

1. The smallest number that is a multiple of both is their ________ ________ ________.

2. The product of a number and any whole number is a ________ of that number.

3. A multiple that two or more numbers share is a ________ multiple.

4. Counting forward by a number other than 1, like 6, 12, 18, is called ________.

5 Writing a number as ____________ numbers multiplied together is its prime factorization.

Write About the Math

How does the LCM help you figure out the fewest hot dogs and buns to buy so they come out even?

¿Cómo te ayuda el mínimo común múltiplo a averiguar la menor cantidad de salchichas y panes que hay que comprar para que salgan parejos?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Least Common Multiple** *Mínimo común múltiplo*
- [ ] **Multiple** *Múltiplo*
- [ ] **Common multiple** *Múltiplo común*
- [ ] **Skip counting** *Conteo salteado*
- [ ] **Prime factorization** *Factorización prima*

Model: *The next shared check is a common multiple of 4 and 6, not their sum.* — Students reason that each astronaut repeats on multiples of their own number, so they only overlap at a common multiple, and adding 4 + 6 does not land on both schedules.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I need to buy ____ hot dogs and ____ buns because the LCM of 8 and 6 is ____.**
- **My answer is ____ because ____.**

  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **My evidence is ____, so ____.**

  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Least Common Multiple
2. **Notes 2:** Multiple
3. **Notes 3:** Common multiple
4. **Notes 4:** Skip counting
5. **Notes 5:** Prime factorization
6. **Watch for this mistake:** *A common mistake in Least Common Multiple is multiplying the two numbers together instead of listing multiples to find the smallest shared one. For example, for 4 and 6, a student calculates $4 \times 6 = 24$ and calls that the LCM, but the actual least common multiple is 12 (multiples of 4: 4, 8, 12; multiples of 6: 6, 12 — 12 is the first number that appears in both lists). This shortcut only works when the two numbers share no common factors (like 4 and 9, where the LCM really is $36 = 4 \times 9$); for numbers like 4 and 6 that share a factor of 2, multiplying overshoots the true LCM. Always list multiples of both numbers and pick the first one that appears in both lists.*
7. **Try It 1:** C. 12 *— Skip count: multiples of 4 are 4, 8, 12; multiples of 6 are 6, 12. The first shared multiple is LCM(4, 6) = 12 minutes.*
8. **Try It 2:** C. 24 *— LCM(8, 12) = 24. Multiples of 8: 8, 16, 24; multiples of 12: 12, 24. The first shared multiple is 24 minutes.*